# Centre-of-Expansion shell cosmology: workflow to

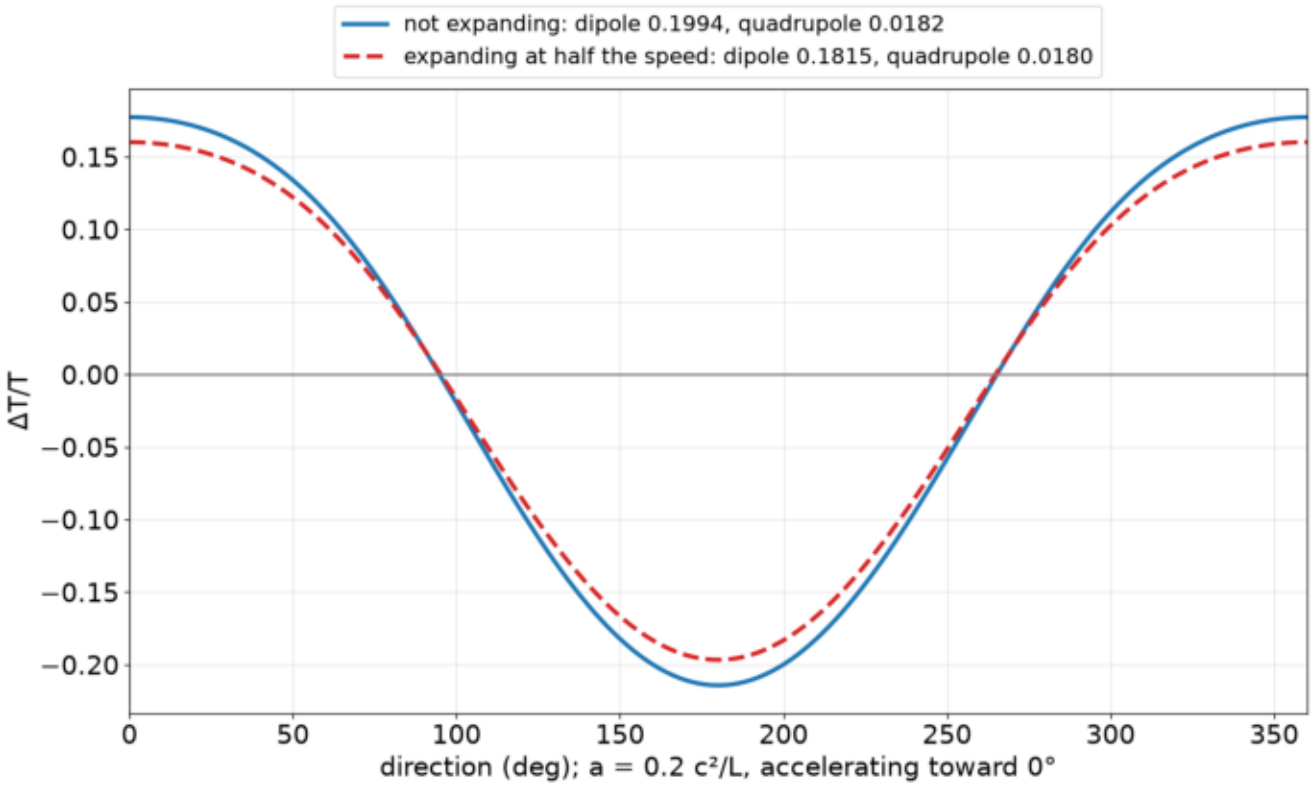
Working writeup for Chris, 19 Oct 2026. It follows the order and terms of Chris's draft 'Centre-of-Expansion shell cosmology' (1 Oct) and the 110-page compilation (7 Oct). Numbers marked [Chris's draft, 1 Oct] or [Chris's 110-page compilation, 7 Oct] are quoted from those documents. Every other number comes from a saved run file under notes/accel, and the file is named in the caption or text.

Abstract. We pose the inverse problem: can non-accelerating expansion in a finite, off-centre geometry look accelerated to an observer who is not at the Centre of Expansion (CoE)? An axisymmetric CoE model with a point core mass fits the Migkas A/A_all cluster anisotropy ($R^2$ 0.810, $\chi^2$ 7.673, 4.78σ over flat). Moving from a point core to shells, and then to a smooth density law locked to the 3% CatWISE gradient, keeps that fit. Placing the model on the sky by Chris's step 3 (+15° tilt, radial gradient, +80.8° roll about the CoE axis) matches the MAX/MIN loci of Migkas Fig. 8. For the CMB, sequential recombination at a fixed radius gives a dipole of 89× Planck. A recombination radius set by a density threshold, and therefore shrinking, brings it to 1.72×, and the mass asymmetry sets a floor of 1.73×. Chris's own results reach the CMB level by a different route: SR Doppler plus the central mass cancel 95.9% of the geometric RMS, and a compensated random web reaches 0.998× [Chris's 110-page compilation, 7 Oct]. Folding that web approach into the new recombination model is the open step.

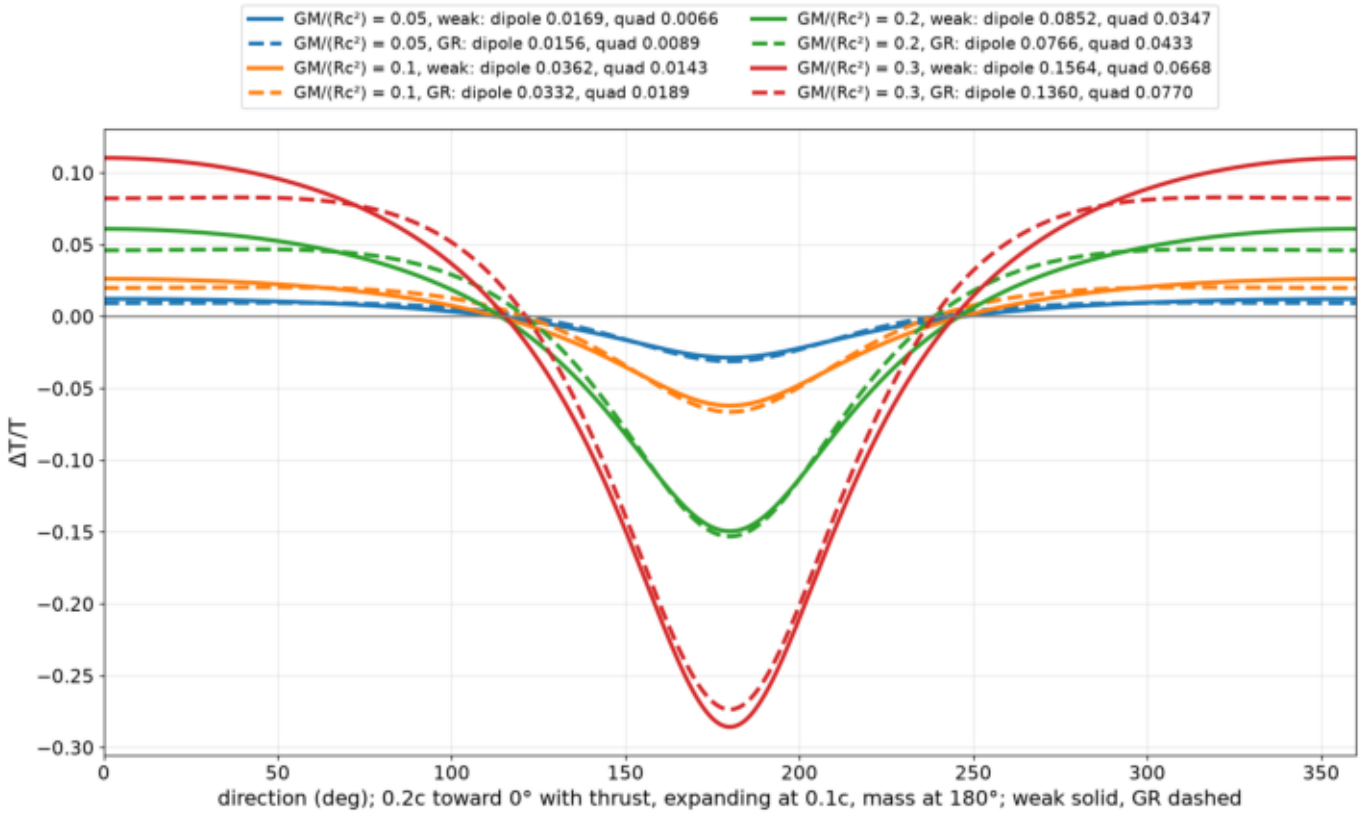
# 1. Can apparent acceleration be explained by simple off-centre ge

The inverse problem [Chris's 110-page compilation, 7 Oct]: Type Ia supernovae show apparent late-time acceleration, but the distance modulus alone does not fix the geometry. Can expansion that is not accelerating, in a finite geometry with an off-centre CoE, look accelerated to us? In this picture, what looks like acceleration is an artefact of sampling the expansion field off-centre, so neither dark energy nor an outward acceleration of the shell is needed. A contracting CoE geometry can look the same on average but differs in its angular signature [Chris's draft, 1 Oct].

Geometry [Chris's draft, 1 Oct]: the local Hubble velocity is free fall toward the CoE minus our own motion. We sit inside a thick ordinary-matter shell at distance D from the CoE (the donut-universe configuration), with a vision radius of about 14.3 Gpc. The figures are the earlier overlays from notes/accel. They illustrate the idea; no fit statistic is attached to them.




Accelerating-case overlay (illustration). [overlay_accel.png]



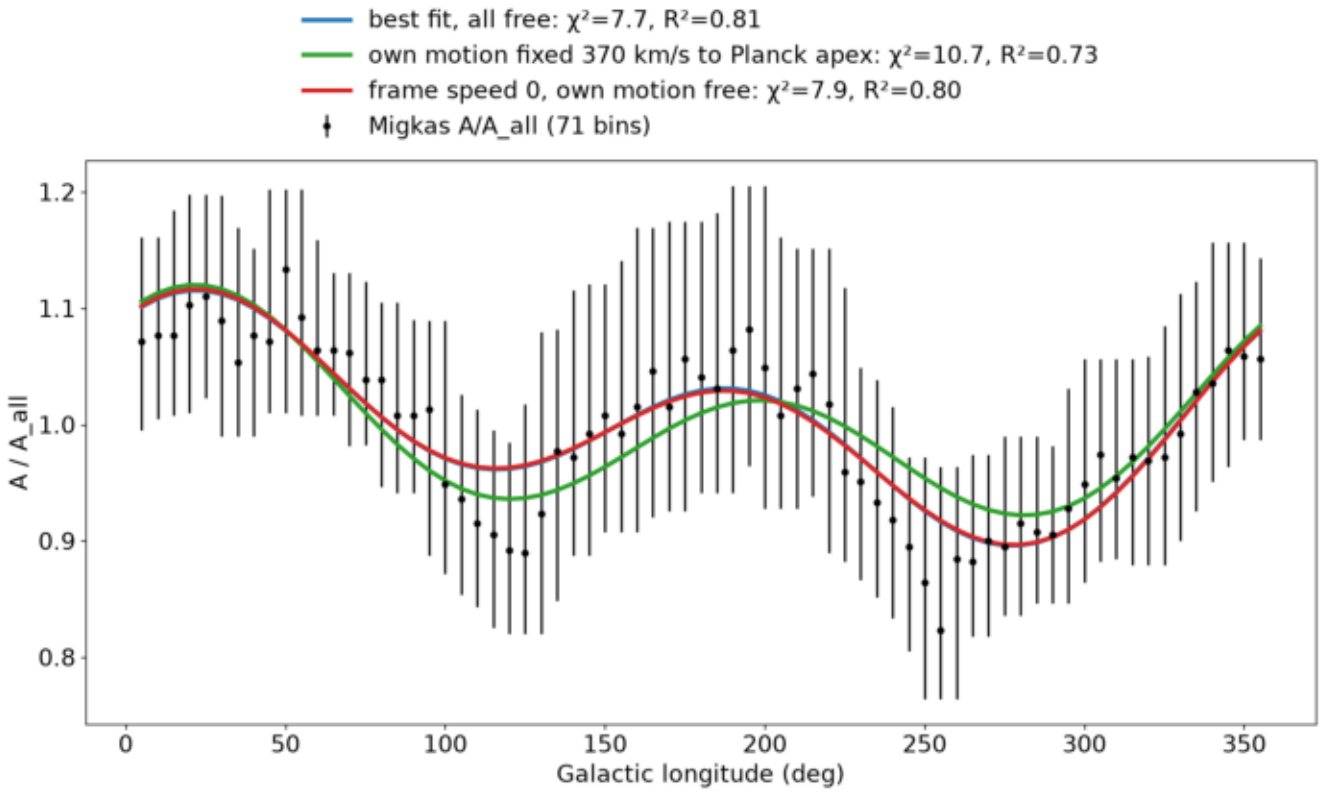

GR constant expansion seen off-centre (illustration). [gr/gr_const_expand.png]

## 2. Axisymmetric point-mass result: fit to Migkas A/A_all

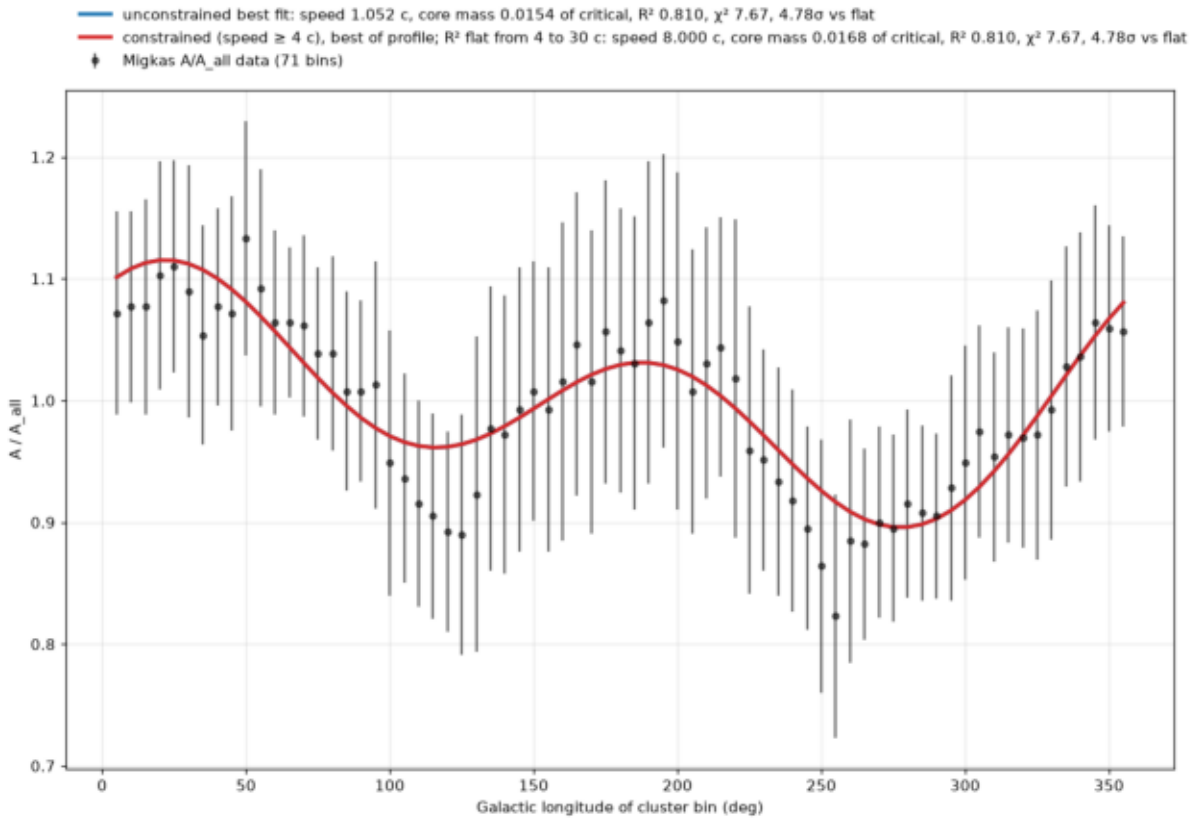
Chris's step 1 compares Migkas A/A_all against Galactic longitude. The axisymmetric model predicts a two-hump pattern about the CoE axis.

Our point-mass fit (point_fit_c4.json and point_fit.json) has seven parameters: core mass, speed, axis longitude, normalisation offset and three peculiar-velocity terms. It gives $\chi^2$ 7.673 on 71 points, $R^2$ 0.810, and $\Delta\chi^2$ 37.0 against a flat line ($\chi^2$ 44.71), which is 4.78σ. The core mass is 0.015 of critical at best fit (median 0.018, 68% range 0.013–0.024) and the axis longitude is 15.6°. Speed is unconstrained: a constrained run at 10c gives $\chi^2$ 7.682, and the profile is flat from about 1c to 30c. The peculiar-velocity amplitude sits on its 3000 km/s bound, which is a limit of the run.

For comparison, Chris's Table 1 Newtonian fit with free shell density gives $R^2$ about 0.909 against Migkas, with a core mass of 0.00265 critical and $\ell 0$ 12.8° [Chris's draft, 1 Oct]. That fit uses a different parameter set (it includes a sine rotation), so the two $R^2$ values do not compare directly.




Point-mass fit to Migkas A/A_all. [gr/three_vel/point/point_fit_migkas.png]

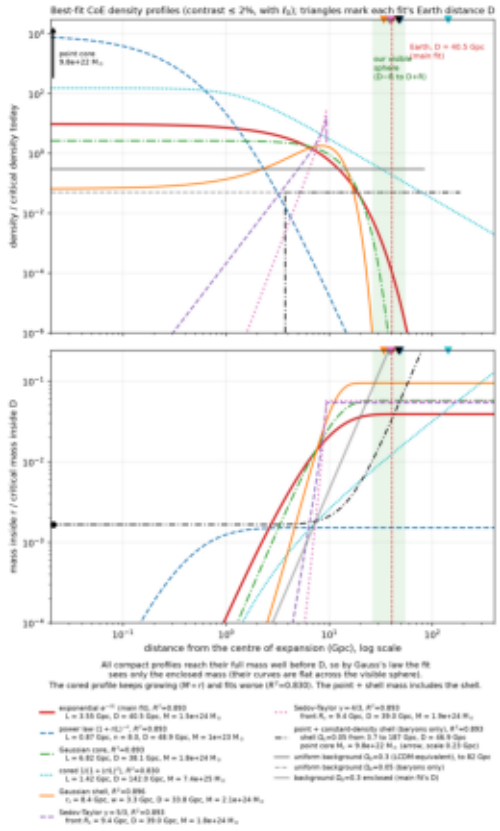



Same fit at 4c. [gr/rings_cmb/v11/migkas_c4_fit.png]

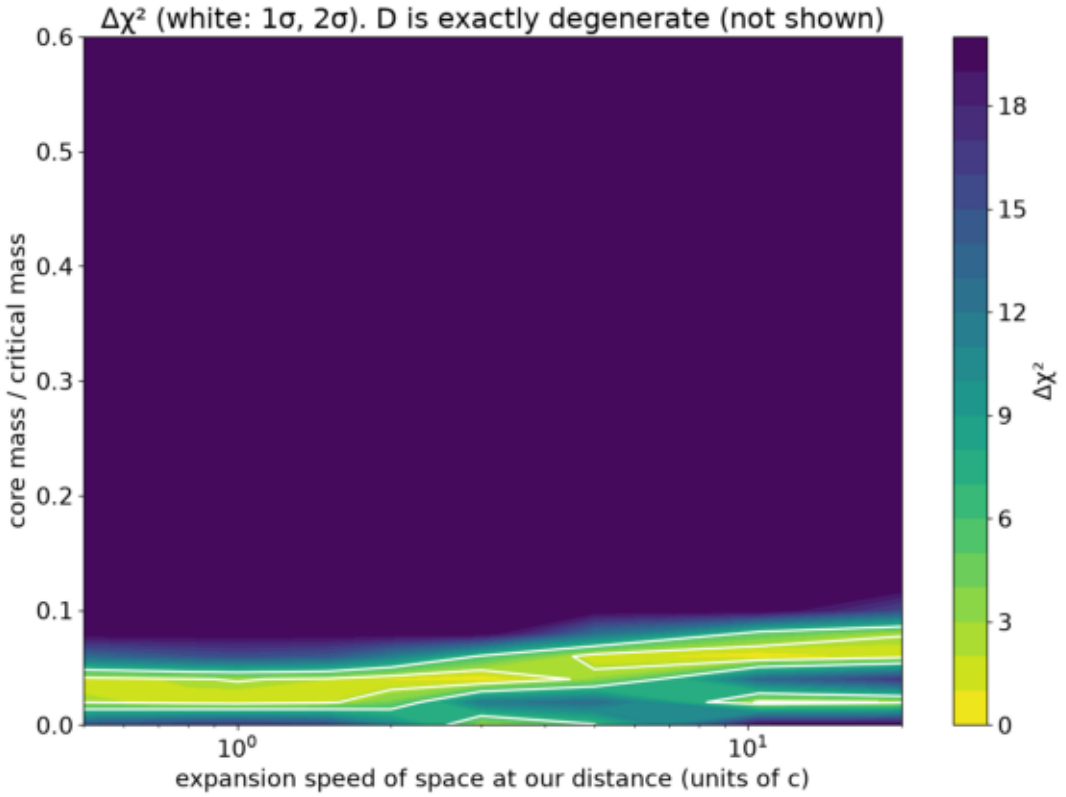
## 3. Shells of increasing mass, then more shells, then a smooth dens

Step 2 of Chris's procedure fits a CoE plus thick shell, both in Newtonian gravity and in exact LTB. Chris's preferred relativistic fit is D ≈ 32.9 Gpc, inner edge ≈ 18.1 Gpc, outer edge ≈ 49.7 Gpc, with the observer mid-shell [Chris's draft, 1 Oct]. In our version, nested shells of increasing mass replace the point core, and the shell count rises until the profile becomes a continuous density. Our physical core-plus-speed grid (phys_fit.json) gives $\chi^2$ 9.90 and $R^2$ 0.745, with a core fraction of 0.04 (1σ 0.04–0.06).

Missing figure: the box has no saved image showing discrete shells, or the step from few shells to many. Below are Chris's density profiles from the spreadsheet model and our core-versus-speed degeneracy plot.




Density profiles, Chris's spreadsheet model.
[/workspace/cosmology/papers/spreadsheet_model/density_profiles.png]

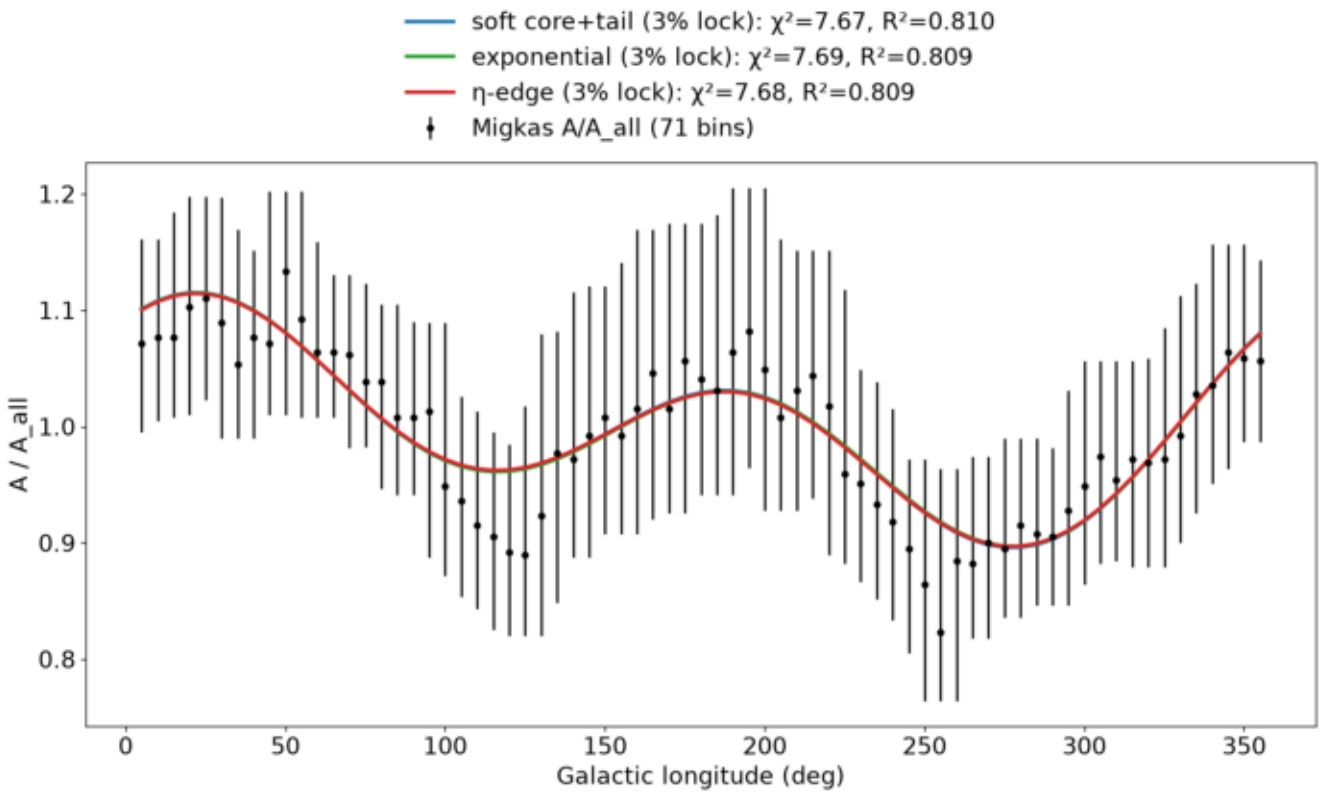



Core mass against speed degeneracy. [gr/three_vel/phys/degeneracy_core_vs_speed.png]

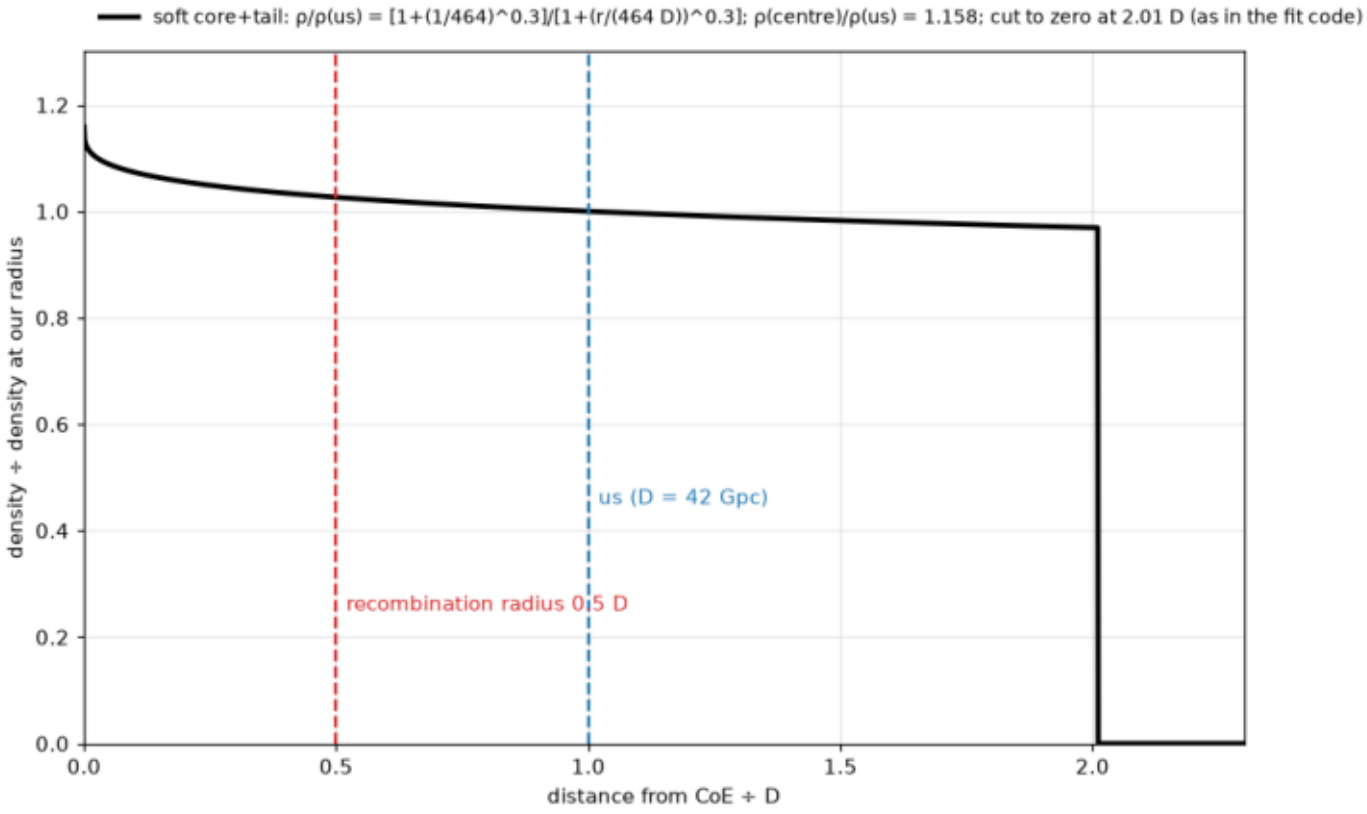
## 4. A family of densities falling off faster or slower outside the visib

We fitted smooth laws that differ only beyond our horizon, each with the outward gradient locked to the 3% CatWISE value. Chris's compilation quotes the CatWISE/Secrest matter dipole as D ≈ 0.0155 [Chris's 110-page compilation, 7 Oct].

The soft core-plus-tail law, ρ ∝ 1/(1+(r/L)^0.3), gives $\chi^2$ 7.674 and $R^2$ 0.810 (fit_soft.json). The η law gives $\chi^2$ 7.678 and $R^2$ 0.809 (fit_eta.json). Migkas cannot tell the laws apart; the CatWISE lock is what sets the slope. In the recombination runs the soft law uses L = 464 D, and the Migkas-fitted mass of 4.36× critical makes the model bound (age 15.8 Gyr). Those two figures come from our earlier runs as reported in conversation; I did not re-read them from a file.




Density-law fits to Migkas. [gr/three_vel/density/density_fit_migkas.png]




Soft core+tail density against distance. [gr/rings_cmb/v11/density_soft_law_4c.png]

## 5. Recombination: the large bubble, with three cases

Chris's step 4: shells expand outward faster than light in a coordinate sense, while the sphere of events visible to us grows at c. Recombination is a spherical surface around the CoE [Chris's draft, 1 Oct]. In our ray pictures each emitter moves radially, and where it crosses the recombination radius (0.5 D) it forms the large bubble whose light comes back to us.

The cases are simultaneous (all emitters recombine at one moment) and sequential (each at its own crossing), each with and without mass (dist_stats.json and earlier runs):

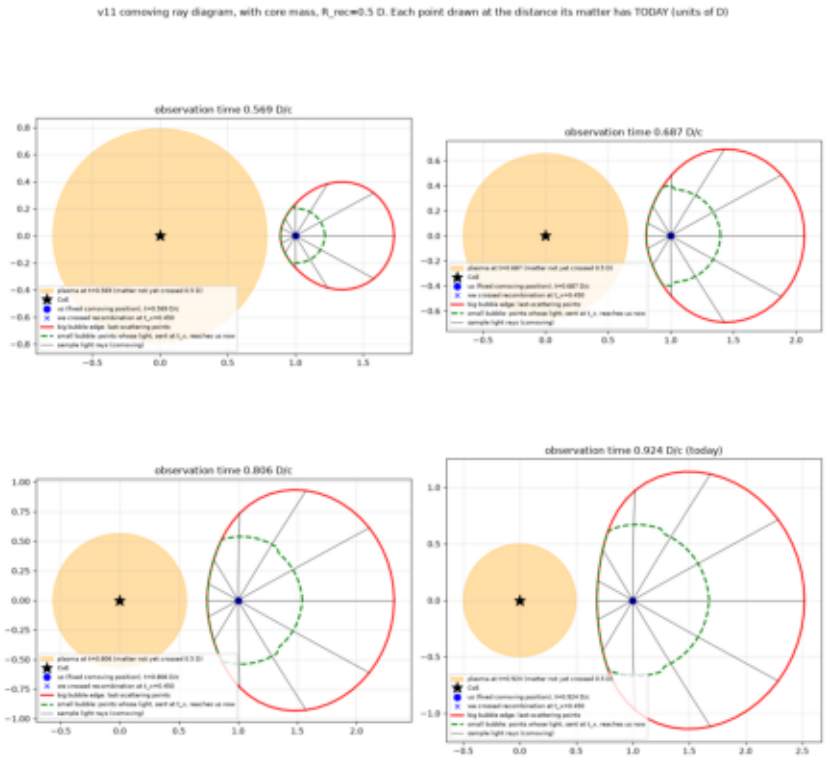
• Simultaneous, no mass: exactly flat, with 1+z = 2.000 everywhere.

• Simultaneous, point mass: 127× Planck.

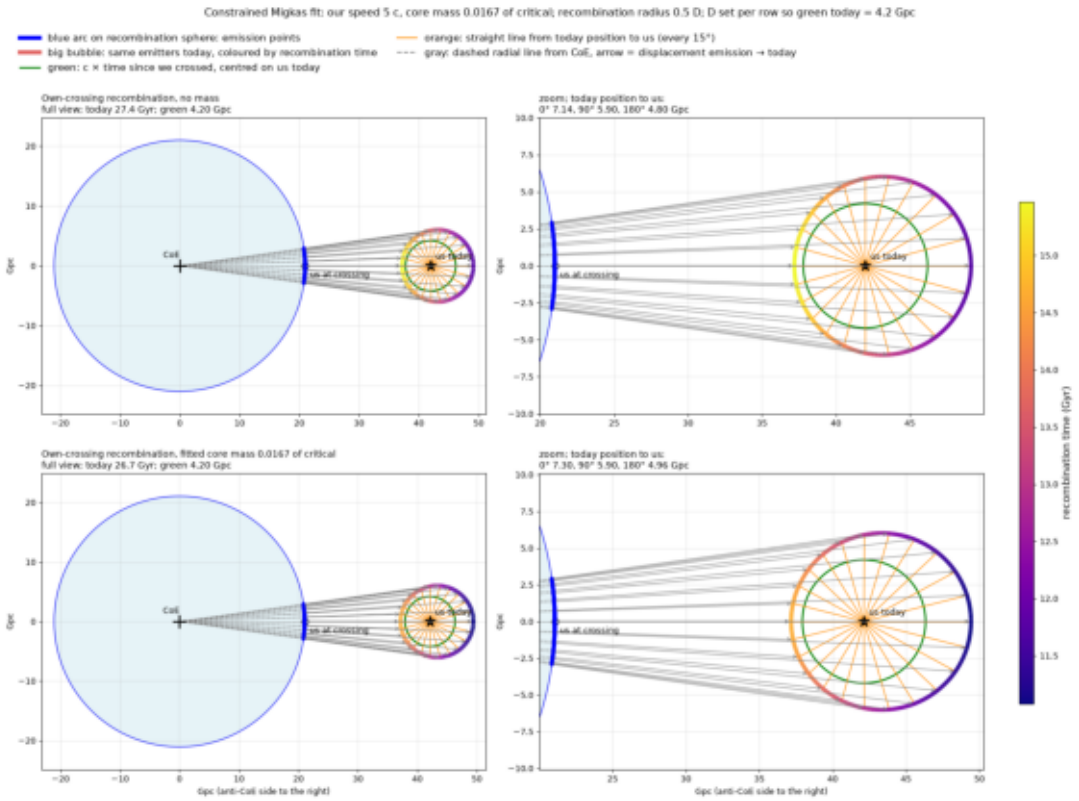
• Sequential, no mass: dipole 0.175.

• Sequential, point mass: dipole 0.200.

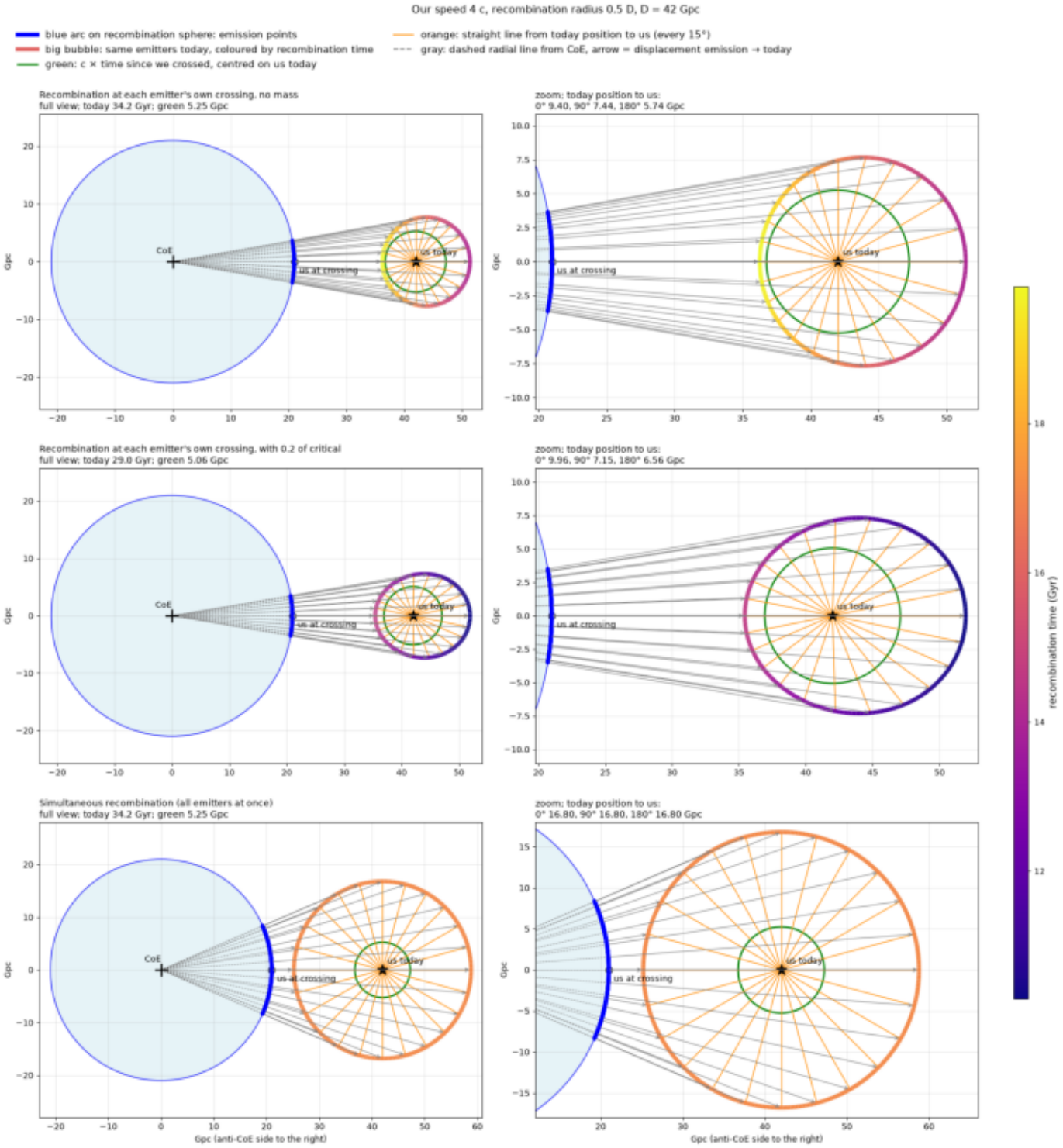
With the distributed soft mass, the leftover after dipole removal is about 1.4–1.9× Planck at equal mass. At the Migkas mass it is 1.37× sequential and 3.44× simultaneous. The sequential dipole itself stays near 0.11–0.17, roughly 100× the observed value. The traced simultaneous row of the 2×3 figure has not yet been redrawn.



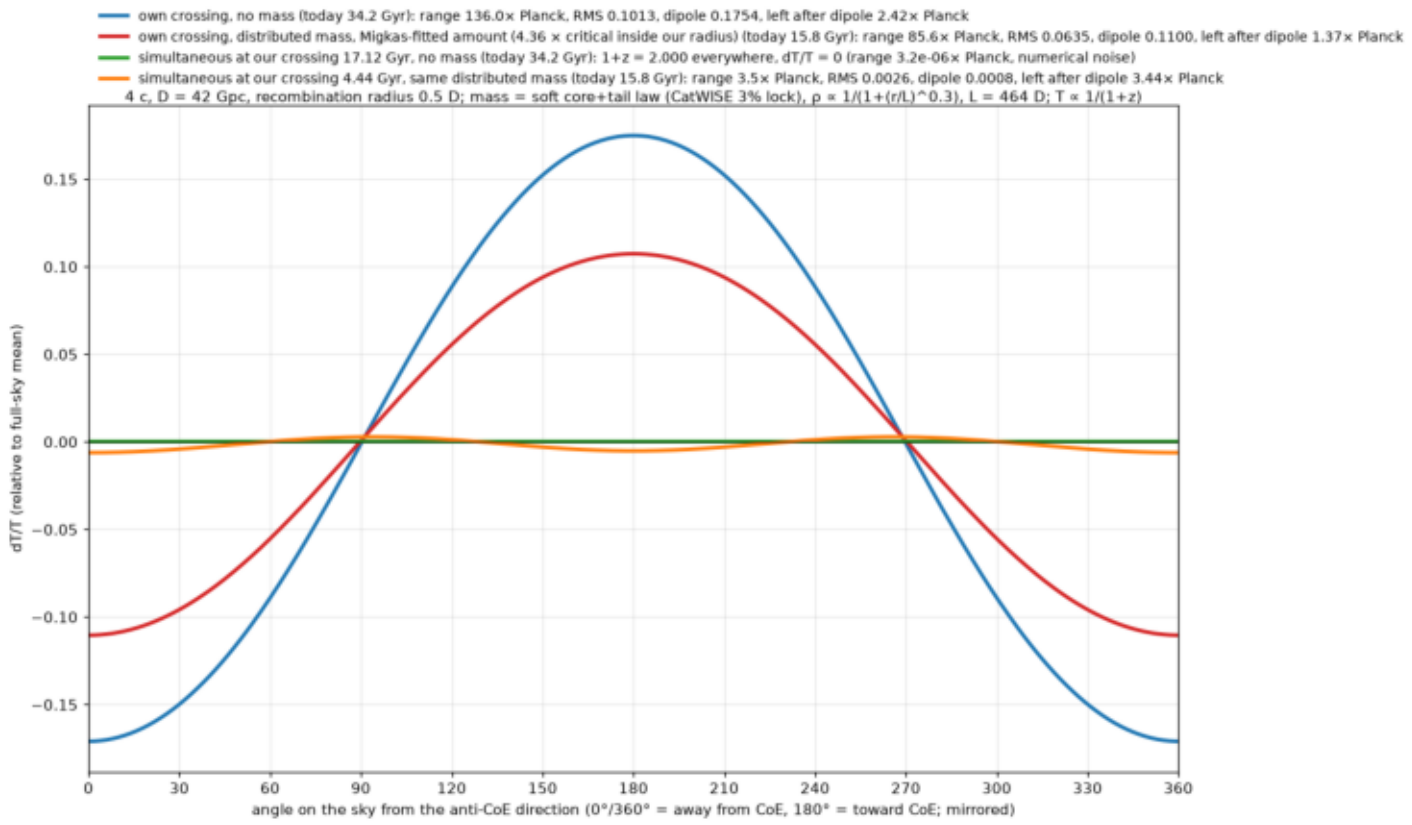

Ray diagram: radial motion, recombination bubble, light returning to us.
[gr/rings_cmb/v11/v11_ray_diagram.png]



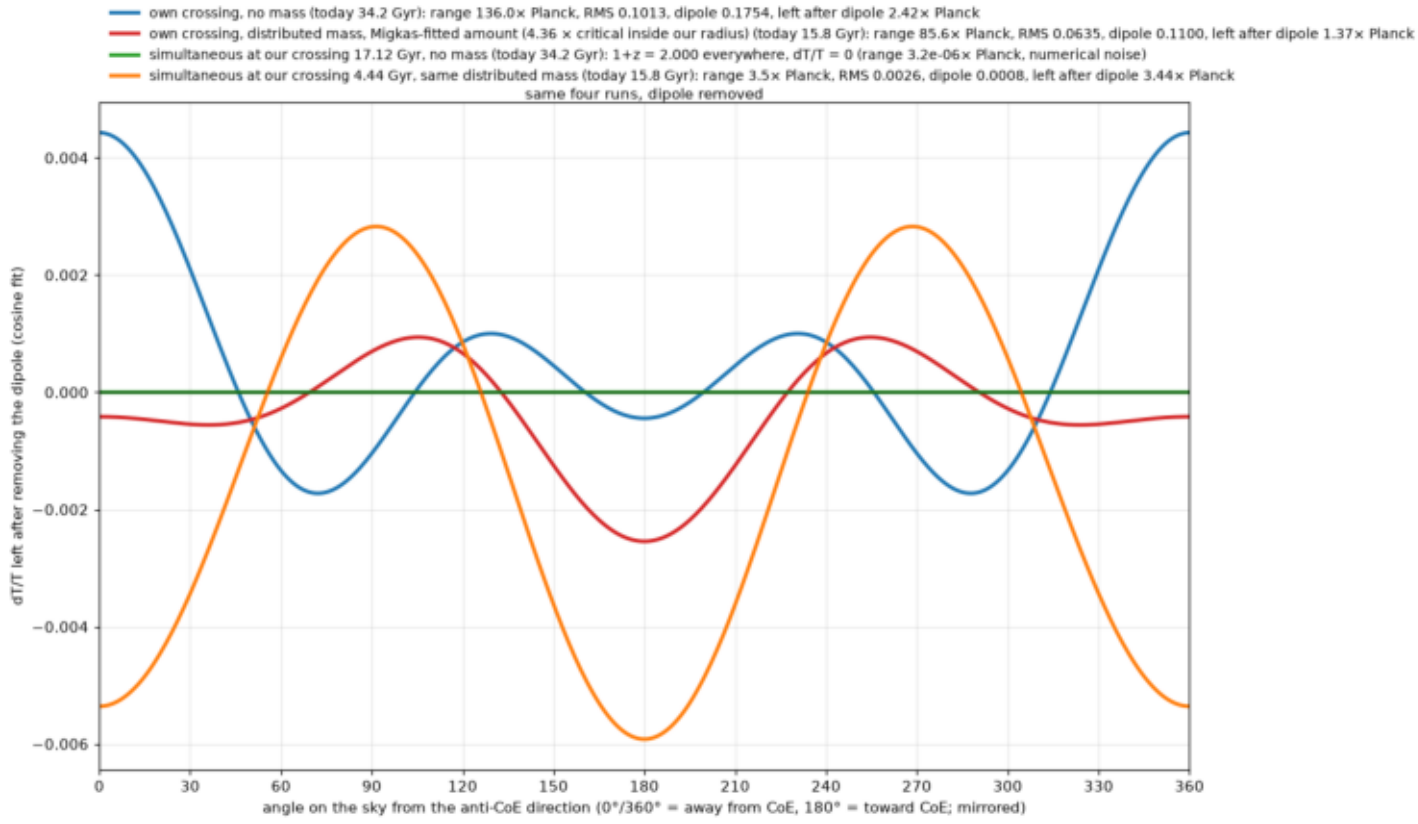

Plain-arc bubble at the Migkas fit. [gr/rings_cmb/v11/migkas_c25_2x2_plainarc.png]




Simultaneous vs sequential, with and without mass. [gr/rings_cmb/v11/emit_cross_4c_2x3_equal_plainarc.png]




dT/T against angle, four cases, distributed mass. [gr/rings_cmb/v11/dTT_4c_4scenarios_distributed.png]



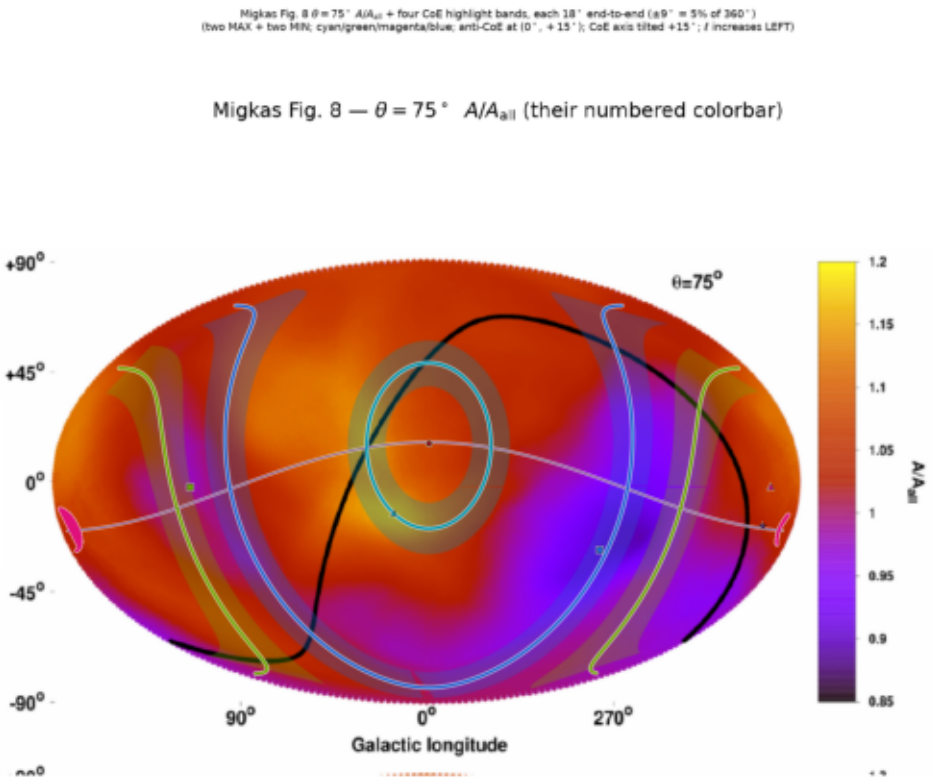

Same after dipole removal. [gr/rings_cmb/v11/dTT_4c_4scenarios_distributed_residual.png]

## 6. Sky orientation (Chris's step 3)

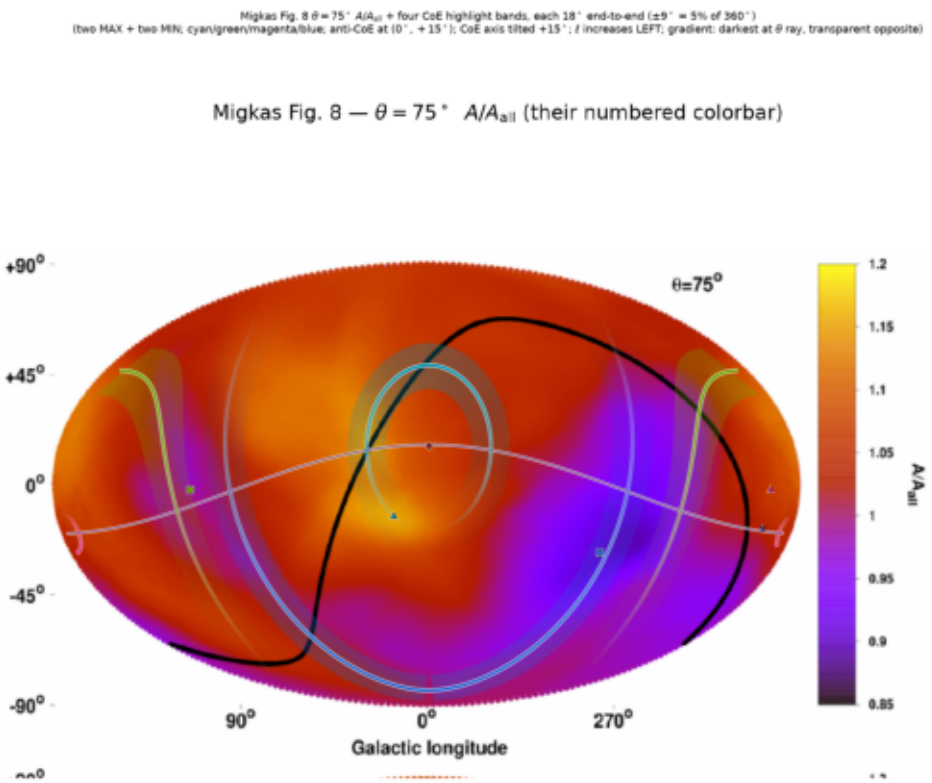
Start from the untilted, no-roll model. Apply a +15° tilt, then a radial gradient, then a +80.8° roll about the CoE axis [Chris's draft, 1 Oct]. Chris's Fig. 7 shows each stage as the model frame, a Mollweide sky map and a 3D view. Chris's Fig. 8 overlays the predicted MAX/MIN bands on Migkas Fig. 8 for tilt alone, tilt plus gradient, and tilt plus gradient plus roll. Each band is 18° end to end. The Migkas MAX markers include (17°, −9°) and (185°, 0°), and the anti-CoE end is near (0°, +15°).

Correction to our earlier runs: roll is a real degree of freedom here. The radial gradient breaks the symmetry about the CoE axis, so rolling about that axis moves the MAX/MIN loci on the sky. Our earlier statement that roll does nothing came from a model without the gradient, and it does not apply to Chris's oriented model. Counting the knobs: tilt, gradient and roll are three orientation choices fitted to Migkas Fig. 8.

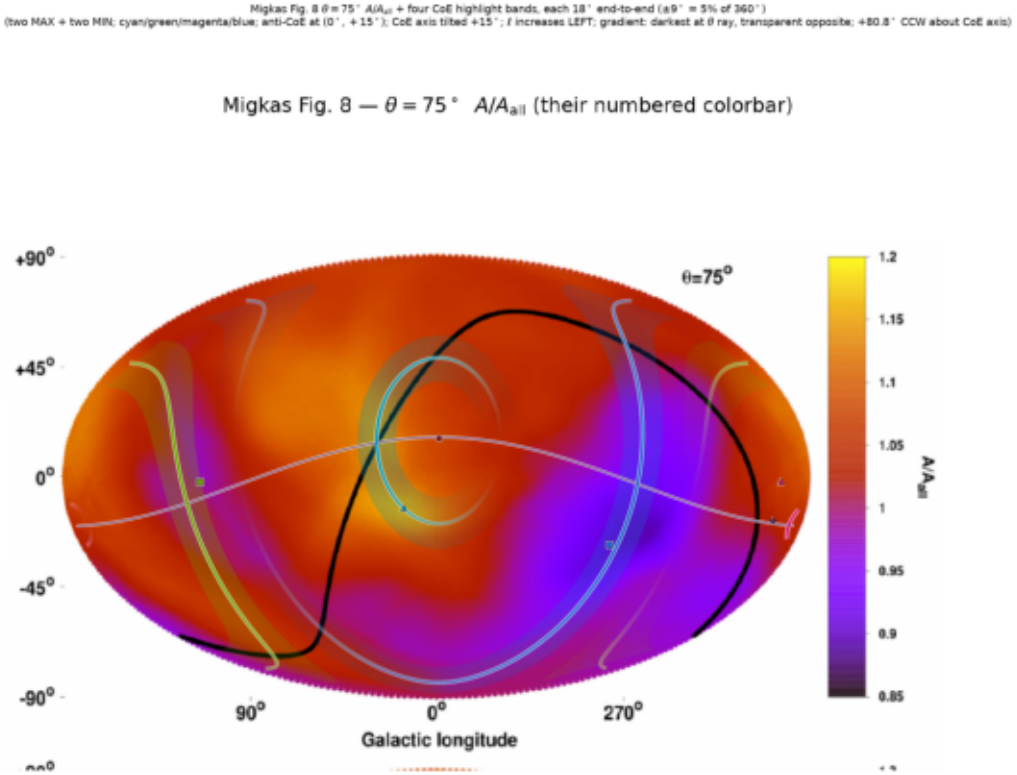
The figures on the next pages are the spreadsheet-model renders on the box (cosmology/papers/spreadsheet_model), which appear to be the source of Chris's Figs. 7–8. One mismatch: the three-panel stack is labelled with the CoE axis tilted +6.6° (anti-CoE at 12.8°, +6.6°), while the overlay sequence and the draft say +15°. Chris should confirm which goes in the paper.



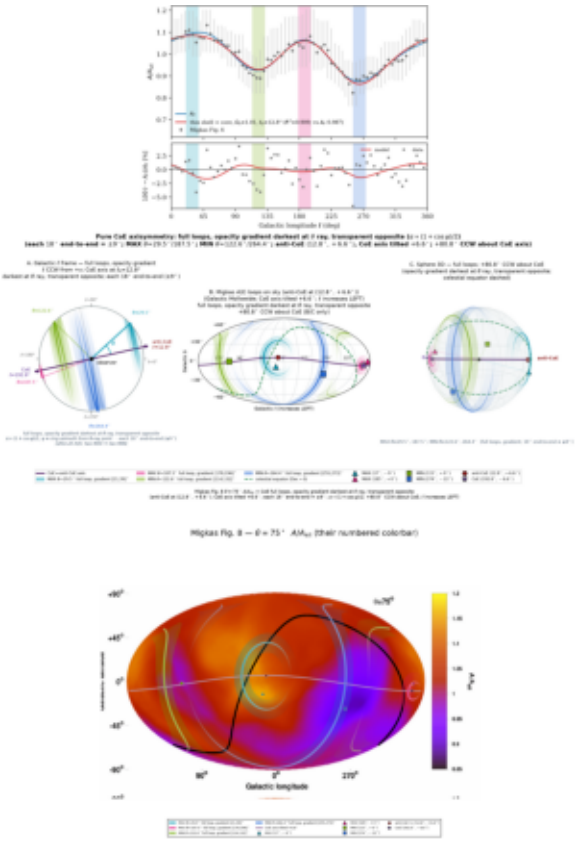

Chris's Fig. 8 sequence, stage 1: +15° tilt only, on Migkas Fig. 8.
[/workspace/cosmology/papers/spreadsheet_model/fig8_overlay_matched.png]




Stage 2: tilt plus radial gradient.
[/workspace/cosmology/papers/spreadsheet_model/fig8_overlay_matched_grad.png]




Stage 3: tilt, gradient and +80.8° roll about the CoE axis.
[/workspace/cosmology/papers/spreadsheet_model/fig8_overlay_matched_grad_rot81.png]



Three-panel stack at +80.8° roll with the A/A_all curve (labelled tilt +6.6°).
[/workspace/cosmology/papers/spreadsheet_model/loop_stack_roll80p8.png]

# 7. The CMB dipole: new runs and Chris's results

• Sequential baseline (dipole_amp.json): 89.2× Planck at mean 1+z ≈ 2. The CoE side recombines about 36% earlier than the anti-CoE side, and this timing difference makes up about 99% of the dipole.

• Simultaneity lever: the dipole falls linearly, through 66.8×, 44.8×, 22.8×, 9.5× and 5.1× as recombination goes from 25% to 95% simultaneous. It reaches 1× when emitters recombine within about 0.14% of one moment. Migkas $\chi^2$ stays at 7.673.

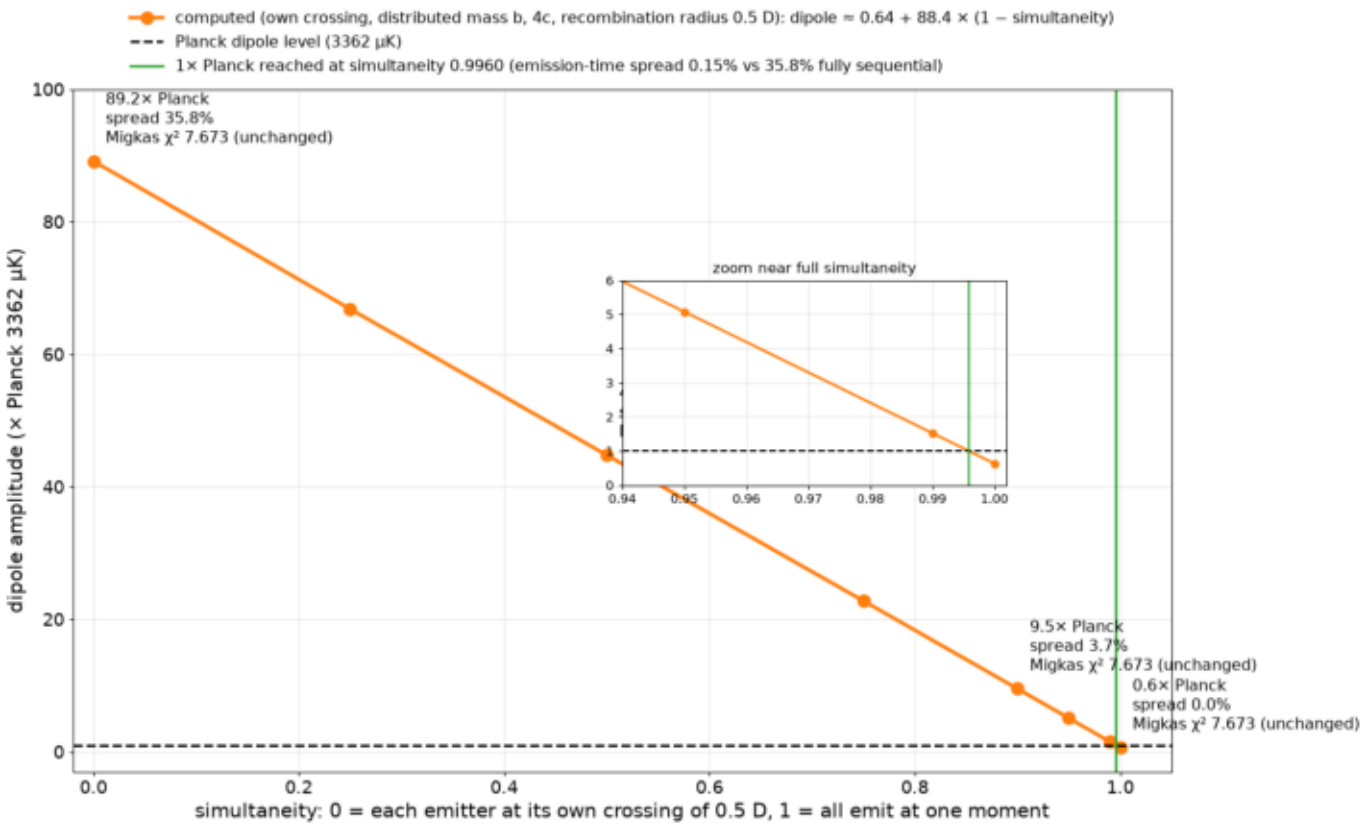
• Shrinking density-threshold radius (shrink_results.json): the sweep rate comes from the CatWISE-locked slope, so it adds no knob. At mean 1+z ≈ 1105 with mass the dipole is 1.72×. Steeper slopes give 24.7×, 78.7× and 235×; the last is the fixed-radius limit.

• Floor (floor_results.json): with mass, the dipole is 0.64× at 0.5 D, 1.26× at 0.2, 1.60× at 0.05 and 1.73× at D/1100. With no mass it is zero at every radius. So 1.73× is the honest minimum, set by the mass asymmetry.

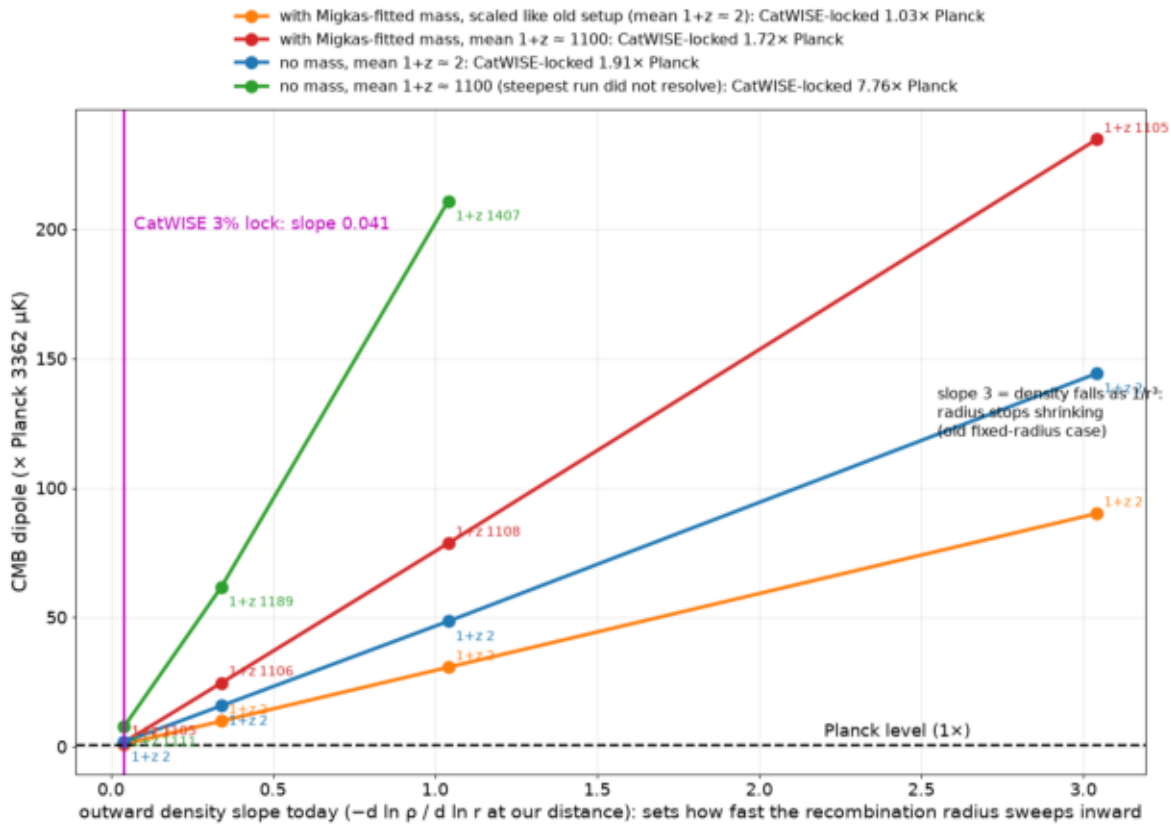
• Adding 370 km/s (motion_results.json): 2.73× with the motion toward the CoE and 0.72× away from it; in both cases the hot end stays at the CoE, 97° from Planck hot. With the motion toward the Planck dipole the result is 1.88×, with the hot end 65° from Planck hot.

Chris's results [Chris's 110-page compilation, 7 Oct]: the pure geometry starts at about 5,750× the 10^-5 level (RMS 5.75e-2). SR Doppler plus central-mass gravitational redshift remove 95.9% of the RMS, leaving about 237× (RMS 2.37e-3, Fix Point 2). A four-clump sketch (four masses of 7.7e51 kg at 24.5 Gpc, $\ell$ = 75°, 162°, 228°, 315°) lowers it to RMS 8.34e-4, about 83×, leaving mainly m = 6 and m = 1. In section IX a compensated random web takes seed 3 from 789× to 0.998×. That run edits 1,099 bodies (12.1%), deletes none and makes no ring. Each void is balanced by its own wall clusters at 1–1.3 void radii. Void sizes follow Jennings–Li–Hu (median 1.5 Gpc), cluster masses follow Tinker (median 1.40e50 kg), and voids fill 29% of the volume. Seeds 2 and 1 stalled at 1.18× and 7.40×.

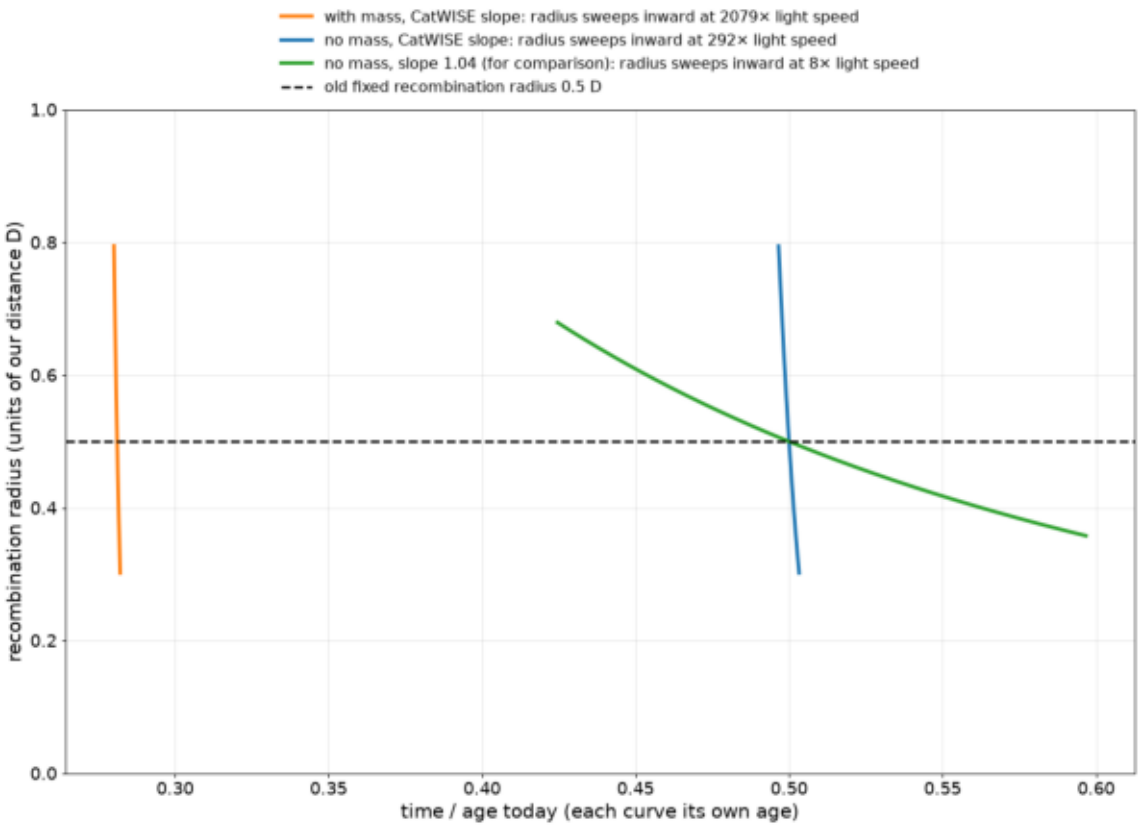
Where the two agree and differ: both find that pure geometry alone overshoots by orders of magnitude and that a physical mechanism removes most of it. Chris's cancellation is Doppler plus central mass on a fixed-radius recombination surface. Ours is near-simultaneous freezing from a shrinking density-threshold radius. The quantities also differ: his are RMS around the ring relative to 10^-5, ours are dipole amplitude relative to Planck's 3362 μK, so 237× and 1.73× cannot be compared directly. Both end with a mass-driven leftover that needs structure, meaning clumps or the web, to remove.



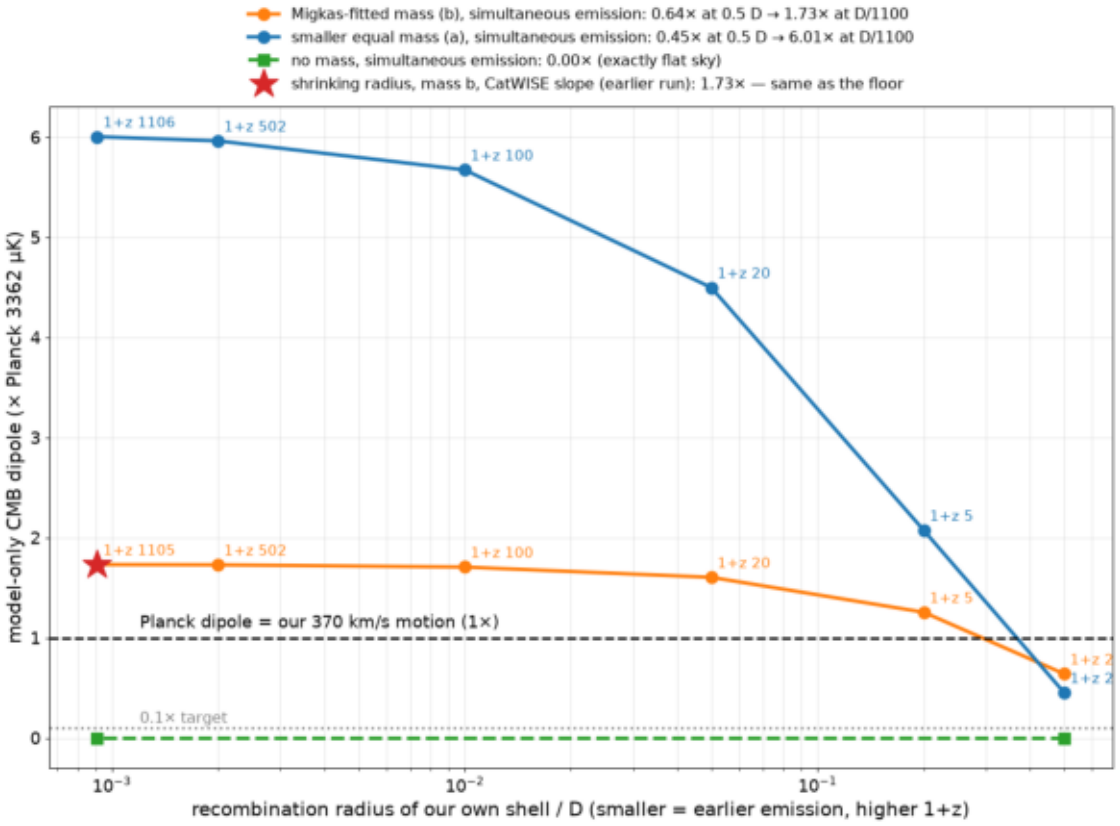

Dipole against degree of simultaneity. [gr/rings_cmb/v11/dipole_amp/dipole_vs_simultaneity.png]



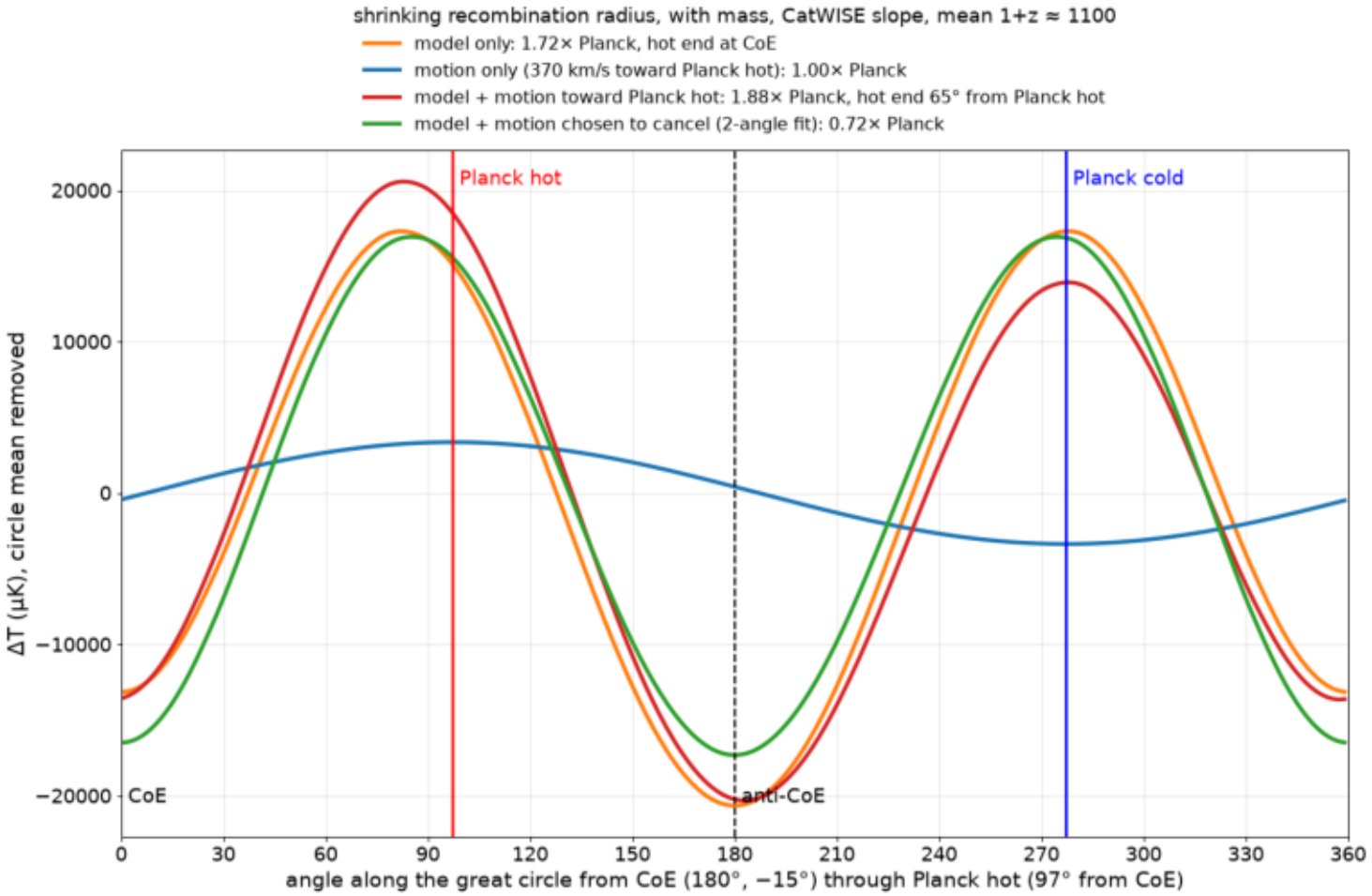

Dipole against density slope (shrink rate). [gr/rings_cmb/v11/shrink_rec/dipole_vs_shrink.png]




Recombination radius against time. [gr/rings_cmb/v11/shrink_rec/Rrec_vs_time.png]




Floor against recombination radius. [gr/rings_cmb/v11/shrink_rec/floor/floor_vs_Rrec.png]




Model dipole combined with 370 km/s motion. [gr/rings_cmb/v11/shrink_rec/motion/dT_combined.png]

## 8. Status and open problems

Fitted so far: Migkas A/A_all (4.78σ over flat), the 3% CatWISE gradient (imposed through the lock), and the Migkas Fig. 8 MAX/MIN loci after Chris's tilt, gradient and roll.

Open: in our recombination model the model's own dipole bottoms out at 1.73× Planck. That is above the roughly 1× from our 370 km/s motion, so it would dominate the observed dipole and point it at the CoE. Our earlier orientation runs had no gradient, so they did not use roll. They should be redone with Chris's radial gradient before any conclusion about the CMB direction.

Route to about 1×: Chris's compensated random web (section IX) shows that rearranging about 12% of a balanced, realistic web flattens his ring residual to 0.998×. Our clusters-and-voids run, which adds such a web on top of the 1.73× smooth dipole, is in progress, with no results yet.

Missing real figures: discrete shells (section 3). Chris's compensated-web figures (Figs. 55–60) are not on the box as image files. They are described in section 7 from the compilation text only.